

Question 1:

a) Evaluate $\int_0^3 \frac{dx}{9+x^2}$ giving your answer in exact form.

1

2

b) Use the table of standard integrals to find the exact value of

2

$$\int_0^4 \frac{dx}{\sqrt{9+x^2}}$$

c) Show that $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan \theta$.

4

Hence express $\tan 67\frac{1}{2}^\circ$ in simplest form.

d) Solve the inequality $x \geq \frac{1}{x}$

4

Question 2:**Marks:**

a) Find the acute angle between the lines $y = \frac{1}{3}x + 3$ and $y = -\frac{2}{3}x + 3$.

2

Give your answer in radians correct to two decimal places.

b) The polynomial $x^3 - 3x + 1 = 0$ has roots α, β and γ . Find the exact value

3

of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

c) i) By using graphs or otherwise show that the curves $y = \ln x$ and $y = 2 - x$ have a point of intersection for which the x co-ordinate is close to 1.5.

1

2

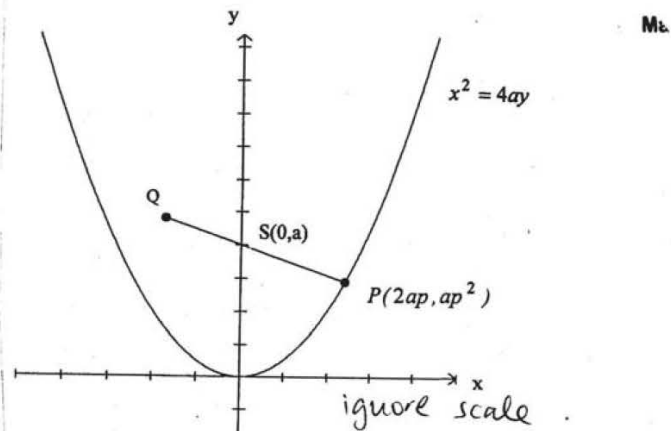
ii) Use $x = 1.5$ and one application of Newton's method to find a better approximation of the x co-ordinate of this point of intersection. Give answer correct to two decimal places.

d) Use the substitution $u = x - 1$ to evaluate $\int_2^5 \frac{x+1}{\sqrt{x-1}} dx$

4

Question 3:

a)



In the diagram above $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$. The point Q lies on PS produced such that Q divides PS externally in the ratio 3:2.

i) Prove that Q has co-ordinates $(-4ap, a(3-2p^2))$ 2

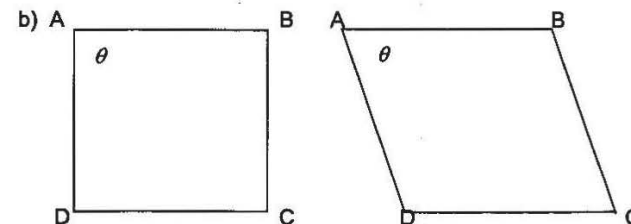
ii) Show that as P varies the locus of Q is another parabola. Find its equation and write down the co-ordinates of its vertex and focus in terms of a. 5

b) Prove by mathematical induction that $\sin(x + n\pi) = (-1)^n \sin x$ where n is a positive integer. 5

Question 4:

Marks:

a) Solve for x: $\log_{\frac{1}{2}}\left(\frac{1}{x}\right) \geq \log_2(3x-1)$ 3



A square ABCD of side 1 unit is gradually 'pushed over' to become a rhombus. The angle at A (θ) decreases at a constant rate of 0.1° / second.

i) Show that the area of the rhombus is equal to $\sin \theta$ 1

ii) At what rate is the area of the rhombus ABCD decreasing when 2

$\theta = \frac{\pi}{6}$? (Give answer correct to 2 decimal places).

iii) At what rate is the shorter diagonal of the rhombus ABCD 3

decreasing when $\theta = \frac{\pi}{3}$? (Give answer correct to 2 decimal places).

c) Prove that $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = 2$ (for $\sin \theta \neq 0$, $\cos \theta \neq 0$) 3

Question 5:

a)

- i) Express $\sqrt{3} \cos 2t - \sin 2t$ in the form $R \cos(2t + \alpha)$ where 2

$$0 < \alpha < \frac{\pi}{2}$$

- ii) Hence or otherwise find all positive solutions of 3

$$\sqrt{3} \cos 2t - \sin 2t = 0$$

b) A particle moves in a straight line and is x metres from a fixed point O after t seconds where:

$$x = 5 + \sqrt{3} \cos 2t - \sin 2t$$

- i) Prove that the acceleration of the particle is $-4(x-5)$. 3

- ii) Between which two points does the particle oscillate. 2
(You may use your answers from part (a))

- iii) At what times does the particle first pass through the point $x=5$. 2

Question 6:

Marks:

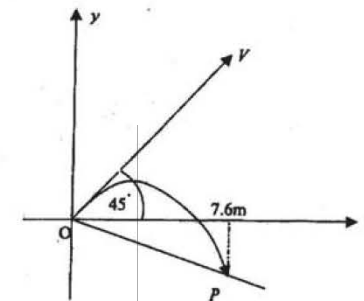
- a) The acceleration at any time t of a body moving in a straight line is $-e^{-2x}$.

When $t=0$, $x=0$, and $v=1$.

- i) Express its velocity v in terms of x . 3

- ii) Express its displacement x in terms of time t . 3

- b) A garden hose placed at the top of an incline releases a stream of water with a velocity of 8 m/s at an angle of 45° with the horizontal. Assuming that $x = vt \cos \alpha$ and $y = vt \sin \alpha - \frac{1}{2}gt^2$ where x and y are the horizontal and vertical displacements of the stream of water from O at any time, $g = 10\text{ m/s}^2$ and the coordinate axes are taken as shown.



- i) Show that the equation of the path of the stream of water is given 3

$$\text{by } y = x - \frac{5x^2}{32}$$

- ii) If the stream of water strikes the incline at the point P , 7.6 m horizontally from O , find the equation of the incline. 3

Question 7:

- a) During the early summer months the rate of increase of the population P of fruit flies is proportional to the excess of the population over 3000.

$$\frac{dP}{dt} = k(P - 3000) \text{ where } k \text{ is a constant. At the beginning of summer the}$$

population is 4000 and 1 month later it is 10 000.

- i) Show that $P = 3000 + Ae^{kt}$ is a solution of the differential equation, 1
A is a constant.

- ii) Find the value of A and that of k. 2

- iii) Find to the nearest 100, the population after $2\frac{1}{2}$ months. 1

- iv) After how many weeks does the population reach $\frac{1}{2}$ million? 1
(Give your answer to 1 decimal place).

- b) Consider the function $y = x^3 e^{-x}$

- i) State the greatest possible domain of the function. 1

- ii) Find the maximum value of the function in the domain. 2

- iii) Show that there are 3 points of inflexion and that one of them has a 3
horizontal tangent.

- iv) Sketch the curve for $-1 \leq x \leq 6$ 1

END OF EXAMINATION

TRIAL HSC 2003 - Extension One SOLUTIONS

COMMENTS

Question 1.

$$a) \int_0^3 \frac{dx}{9+x^2} = \frac{1}{3} \tan^{-1} \frac{x}{3} \Big|_0^3 \checkmark$$

$$= \frac{1}{3} \tan^{-1} 1 - \frac{1}{3} \tan^{-1} 0$$

$$= \frac{1}{3} \times \frac{\pi}{4}$$

$$= \frac{\pi}{12} \checkmark$$

$$b) \int_0^4 \frac{dx}{\sqrt{9+x^2}} = \ln(x + \sqrt{x^2+9}) \Big|_0^4 \checkmark$$

$$= \ln 9 - \ln 3$$

$$= \ln 3 \checkmark$$

$$c) \frac{1 - \cos 2\theta}{\sin 2\theta} = \frac{1 - (1 - 2\sin^2 \theta)}{2\sin \theta \cos \theta} \checkmark$$

$$= \frac{2\sin^2 \theta}{2\sin \theta \cos \theta}$$

$$= \tan \theta \checkmark$$

$$\tan 67\frac{1}{2}^\circ = \frac{1 - \cos 135^\circ}{\sin 135^\circ}$$

$$= \frac{1 + \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} \checkmark$$

$$= \frac{1}{\frac{1}{\sqrt{2}}} \checkmark$$

$$= \sqrt{2} + 1$$

• some extension II students made this a difficult integral.

• some forgot the $\frac{1}{3}$ at the front
• answer needs to be in degrees

• Some students did not use table of integral
• another error
 $I = \ln(3 + \sqrt{1+x^2})$

• generally well done.

• some need to revise exact values.

$$1d) x \geq \frac{1}{x} \quad x \neq 0.$$

$$x^3 \geq x$$

$$x^3 - x \geq 0 \checkmark$$

$$x(x^2 - 1) \geq 0$$

$$x(x-1)(x+1) \geq 0 \checkmark$$

$$-1 \leq x < 0 \text{ or } x \geq 1 \checkmark$$

OR

$$\text{If } x > 0$$

$$x^2 \geq 1 \checkmark$$

$$\therefore x \geq 1 \text{ or } x \leq -1$$

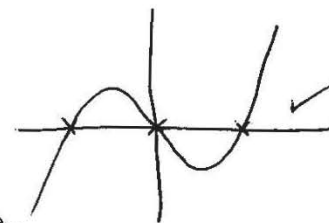
$$\therefore \text{Soln } x \geq 1 \checkmark$$

$$\text{If } x < 0$$

$$x^2 \leq 1 \checkmark$$

$$-1 \leq x \leq 1.$$

$$\therefore \text{Soln } -1 \leq x < 0 \checkmark$$



COMMENT

• generally poorly done.

Need to solve by

a) examining critical pts or

b) mult. bs. by x^2 .

or

c) use two cases - give 2 partial solns and then an overall soln.

SOLUTIONS

COMMENTS

Question 2.

$$a) m_1 = \frac{1}{3} \quad m_2 = -\frac{2}{3}$$

$$\tan \theta = \frac{\frac{1}{3} + \frac{2}{3}}{1 + \frac{1}{3} \times -\frac{2}{3}}$$

$$= \frac{1}{1 - \frac{2}{9}} \checkmark$$

$$= \frac{9}{7} \checkmark$$

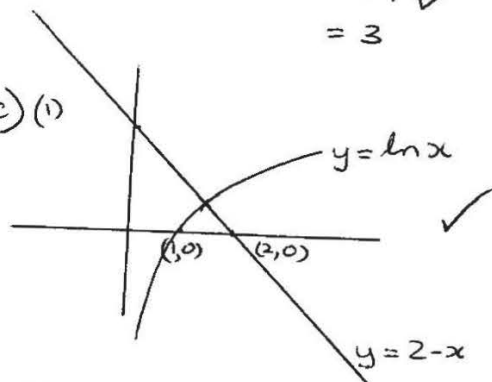
$$\theta = 0.91^\circ \text{ to two decimal places. } \checkmark$$

$$b) \frac{1}{2} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \checkmark$$

$$= \frac{-3}{-1} \checkmark$$

$$= 3$$

c) (i)



Intersection point close to $x = 1.5$

$$(ii) P(x) = \ln x - 2 + x$$

$$P'(x) = \frac{1}{x} + 1$$

$$\therefore x_2 = 1.5 - \left(\frac{\ln 1.5 - 0.5}{1.7/3} \right) \checkmark$$

$$= 1.56 \text{ correct to 2 dec. places } \checkmark$$

$$d) \int_2^5 \frac{x+1}{\sqrt{x-1}} dx$$

$$u = x-1$$

$$du = dx$$

$$= \int_1^4 \frac{u+2}{\sqrt{u}} du$$

① correct substitution ✓

① correct limits

$$= \int_1^4 (\sqrt{u} + 2u^{-1/2}) du \checkmark$$

$$= \left[\frac{2}{3} u^{3/2} + 4u^{1/2} \right]_1^4 \checkmark$$

$$= \left(\frac{16}{3} + 8 \right) - \left(\frac{2}{3} + 4 \right)$$

$$= 8\frac{2}{3} \checkmark$$

No penalty for small arithmetic error to get $8\frac{2}{3}$

$-\frac{1}{2}$ for error finding approximate value of $\tan^{-1} \frac{9}{7}$. No penalty if in degrees.

SOLUTIONS.

Question 3

a) (i) $P(2ap, ap^2) \quad S(0, a)$

$$\therefore Q = \left(\frac{-2 \times 2ap + 0}{1}, \frac{-2 \times ap^2 + 3a}{1} \right)$$

$$= (-4ap, a(3 - 2p^2))$$

(ii) From $x = -4ap$

$$p = \frac{x}{-4a}$$

$$y = a(3 - 2\left(\frac{x^2}{16a^2}\right))$$

$$= 3a - \frac{x^2}{8a}$$

$$\frac{x^2}{8a} = -y + 3a$$

$$x^2 = -8a(y - 3a)$$

This is another parabola

Vertex $(0, 3a)$

Focal length is $2a$. $\therefore$ Focus $(0, a)$

b) If $n=1$. $\sin(x+\pi) = \sin x \cos \pi + \cos x \sin \pi$

$$= -\sin x + 0$$

$$= (-1)^1 \sin x$$

$\therefore$ True for $n=1$

COMMENTS

Assume true for $n=k$

$$\text{i.e. } \sin(x+k\pi) = (-1)^k \sin x$$

consider $n=k+1$

$$\sin(x+(k+1)\pi) = \sin(x+k\pi+\pi)$$

$$= \sin(x+k\pi) \cos \pi + \cos(x+k\pi) \sin \pi$$

$$= (-1)^k \sin x \times -1 + 0$$

$$= (-1)^{k+1} \sin x$$

which is of same form as for $n=k$.

$\therefore$ If true for $n=k$, it is also true for $n=k+1$. Since it is true for $n=1$, it is true for $n=2$ and hence all following positive integers.

One mark for both Step 1 and Step 2.

3 marks for Step 3

Any variation of this quadratic was marked correct.

Many students did not find focal length
 $4a = -8a, l = -2a$

SOLUTIONS

COMMENTS

Question 4.

$$a) \log_{\frac{1}{2}} x \geq \log_2 (3x-1)$$

$$\frac{\log_2 \frac{1}{2} x}{\log_2 \frac{1}{2}} \geq \log_2 (3x-1) \checkmark$$

$$\frac{\log_2 \frac{1}{2} x}{-1} \geq \log_2 (3x-1)$$

$$\log_2 x \geq \log_2 (3x-1) \checkmark$$

$$x \geq 3x-1$$

$$2x-1 \leq 0$$

$$x \leq \frac{1}{2}$$

$$\text{But } x > \frac{1}{3} \text{ since } 3x-1 > 0.$$

$$\therefore \text{Soln } \frac{1}{3} \leq x \leq \frac{1}{2} \checkmark$$

$$b) i) \text{Area} = 2 \times \left(\frac{1}{2} \times 1 \times 1 \times \sin \theta \right) = \sin \theta \checkmark$$

$$ii) \frac{dA}{dt} = \frac{dA}{d\theta} \frac{d\theta}{dt} = \cos \theta \times -0.1 \checkmark$$

$$\text{when } \theta = \frac{\pi}{6} \frac{dA}{dt} = \frac{\sqrt{3}}{2} \times -0.1 = -0.09 \text{ m}^2/\text{s} \checkmark$$

Many Problems
Not Many people realised

$$3x-1 > 0$$

$$\therefore x > \frac{1}{3}$$

Straight away

3

Good

Ok

Substitution caused problems.

$$(iii) l^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \times \cos \theta = 2 - 2 \cos \theta$$

$$l = \sqrt{2} (1 - \cos \theta)^{1/2} \checkmark$$

Ok

$$\frac{dl}{dt} = \frac{dl}{d\theta} \frac{d\theta}{dt}$$

$$= \left(\sqrt{2} \times \frac{1}{2} \times (1 - \cos \theta)^{-1/2} \times \sin \theta \right) \times -0.1 \checkmark$$

Differentiation of l caused problems

$$\text{At } \theta = \frac{\pi}{3} \frac{dl}{dt} = \sqrt{2} \times \frac{1}{2} \times \sqrt{2} \times \frac{\sqrt{3}}{2} \times -0.1 = -0.09 \text{ m/s} \checkmark$$

A lot of people left off the $\sin \theta$ from chain rule

$$c) \frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta}$$

6

$$= \frac{\sin \theta \cos 2\theta + \cos \theta \sin 2\theta}{\sin \theta} - \frac{\cos \theta \cos 2\theta - \sin \theta \sin 2\theta}{\cos \theta} \checkmark$$

$$= \cos 2\theta + \frac{\cos \theta}{\sin \theta} (\sin 2\theta) - \cos 2\theta + \frac{\sin \theta}{\cos \theta} \sin 2\theta \checkmark$$

$$= 2 \sin \theta \cos \theta \left(\frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cos \theta} \right)$$

$$= 2.$$

Well done Overall

3

SOLUTIONS

COMMENTS

Question 5.

$$a) (i) \sqrt{3} \cos 2t - \sin 2t = 2 \cos(2t + \alpha)$$

$$\left[\begin{aligned} \tan \alpha &= \frac{1}{\sqrt{3}} \\ \alpha &= \frac{\pi}{6} \end{aligned} \right]$$

$$= 2 \cos(2t + \frac{\pi}{6})$$

$$(ii) \cos(2t + \frac{\pi}{6}) = 0$$

$$2t + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$2t = \frac{\pi}{3}, \frac{4\pi}{3}, \frac{7\pi}{3}, \dots$$

$$t = \frac{\pi}{6}, \frac{4\pi}{6}, \frac{7\pi}{6}, \dots$$

$$= \frac{(3n+1)\pi}{6} \text{ for } n = 0, 1, 2, \dots$$

Many did not know general solution either this way OR

$$2t + \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{2}$$

$$\therefore 2t = 2n\pi - \frac{\pi}{6} \pm \frac{\pi}{2}$$

$$t = n\pi - \frac{\pi}{12} \pm \frac{\pi}{4}$$

$$b) (i) x = 5 + \sqrt{3} \cos 2t - \sin 2t$$

$$\dot{x} = -2\sqrt{3} \sin 2t - 2 \cos 2t$$

$$\ddot{x} = -4\sqrt{3} \cos 2t + 4 \sin 2t$$

$$= -4(\sqrt{3} \cos 2t - \sin 2t)$$

$$= -4(x - 5)$$

Well done.

(ii) Motion is Simple Harmonic Motion Many did not use part (a) centre of motion is 5, amplitude is 2 from part (a).
 $\therefore$ Particle oscillates between 3 and 7 for itself

$$(iii) \text{ At } x=5: 5 = 5 + \sqrt{3} \cos 2t - \sin 2t$$

$$0 = \sqrt{3} \cos 2t - \sin 2t$$

$$\therefore t = \frac{\pi}{6} \text{ (from (ii))}$$

SOLUTIONS

COMMENT

Question 6.

$$a) (i) \ddot{x} = -e^{-2x}$$

$$\frac{d}{dx} \frac{1}{2} v^2 = -e^{-2x}$$

$$\frac{1}{2} v^2 = \frac{-e^{-2x}}{-2} + C$$

$$\frac{v^2}{2} = \frac{e^{-2x}}{2} + C$$

When $v=1, x=0$

$$\frac{1}{2} = \frac{1}{2} + C \therefore C=0$$

$$v^2 = e^{-2x}$$

$$v = \pm \sqrt{e^{-2x}}$$

Since $v=1$ when $x=0$

$$v = \sqrt{e^{-2x}} = e^{-x}$$

$$(ii) v = e^{-x}$$

$$\frac{dx}{dt} = e^{-x}$$

$$\frac{dt}{dx} = e^x$$

$$t = e^x + C$$

When $t=0, x=0 \therefore C=-1$

$$t = e^x - 1$$

$$e^x = t + 1$$

$$x = \ln(t+1)$$

Most students who used $\ddot{x} = \frac{d}{dx}(\frac{1}{2}v^2)$ performed well on this question. However, many did not explain why v is the positive square root.

Some students used $\frac{dx}{dt}$ correctly but did not make x the subject. Some used an incorrect formula from (i) and could not achieve a final result.

SOLUTIONS

COMMENTS

$$a) (i) x = Vt \cos \alpha, \quad y = Vt \sin \alpha - \frac{1}{2}gt^2$$

$$g=10, \quad v=8, \quad \alpha=45^\circ$$

$$\therefore x = 4\sqrt{2}t, \quad y = 4\sqrt{2}t - 5t^2 \quad \checkmark$$

$$y' = 4\sqrt{2} \left(\frac{x}{4\sqrt{2}} \right) - 5 \cdot \frac{x^2}{32} \quad \checkmark$$

$$= x - \frac{5x^2}{32} \quad \checkmark$$

(ii) Equation of line of incline is $y = mx$. $\checkmark$

$$\therefore mx = x - \frac{5x^2}{32}$$

when $x = 7.6$

$$7.6m = 7.6 - \frac{5 \times 7.6^2}{32} \quad \checkmark$$

$$m = 1 - \frac{5 \times 7.6}{32}$$

$$= -\frac{3}{16}$$

$$\therefore y = -\frac{3x}{16} \text{ is required equation.} \quad \checkmark$$

$$= 0.1875x$$

Generally performed well

Some students made this question much more complicated than it was meant to be

QUESTION 7: SOLUTIONS

COMMENTS

$$a) (i) P = 3000 + Ae^{kt} \quad \text{so } Ae^{kt} = P - 3000$$

$$\frac{dP}{dt} = kAe^{kt}$$

$$= k(P - 3000) \quad \checkmark$$

$$(ii) t=0, \quad P=4000$$

$$4000 = 3000 + Ae^0$$

$$\therefore A = 1000 \quad \checkmark$$

$$t=1, \quad P=10,000$$

$$\therefore 7000 = 1000e^k$$

$$e^k = 7$$

$$k = \ln 7 \quad \checkmark$$

$$\approx 1.946 \text{ to 3 dec places.}$$

$$(iii) t=2.5$$

$$P = 3000 + 1000e^{2.5k}$$

$$= 132600 \text{ to nearest 100}$$

$$(iv) 500,000 = 3000 + 1000e^{kt}$$

$$497000 = 1000e^{kt}$$

$$e^{kt} = 497$$

$$kt = \ln 497$$

$$t = 3.19 \dots \text{ months}$$

$$= 12.8 \text{ weeks}$$

correct to 1 dec. place.

many students were unclear on their starting point, and what substitutions they were making.

many could not convert months to weeks! (and thus lost the mark)

SOLUTIONS

COMMENTS

b) (i) $D = \{x : x \in \mathbb{R}\}$. ✓

(ii) $y' = -x^3 e^{-x} + 3x^2 e^{-x}$
 $= x^2 e^{-x} (-x + 3)$
 $= 0 \text{ if } x = 0 \text{ or } x = 3.$

$y'' = x^3 e^{-x} + e^{-x} \cdot 3x^2 + -3x^2 e^{-x}$
 $+ 6x e^{-x}$
 $= x e^x (x^2 - 6x + 6)$ ✓

At $x = 0$ $y'' = 0$ $\therefore$ horizontal inflection

$x = 3$ $y'' < 0$ $\therefore$ max at $x = 3$
 Maximum value is $3^3 e^{-3} = \frac{27}{e^3}$ ✓

① showing $x = 3$ is a max (any method) - this was often left out.

① max value - many did not read the question and left this out.

(iii) Possible pts of inflection at

$y'' = 0$ at $x = 0$ or $x = \frac{6 \pm \sqrt{12}}{2}$
 $= 3 \pm \sqrt{3}$. ✓

① 3 points from $\frac{dy''}{dx} = 0$

① 3 points are correct

At $x = 0$ horizontal inflection

Since changes concavity at each point - all are points of inflection

① show $x = 0$ is horizontal $\Rightarrow$ indicate $y' = 0$.

many lost the graph mark because horizontal inflection at $x = 0$ was not clear!

